

SARDER JUNAID AHMED

PhD Applicant in Computational Science
Machine Learning Researcher — Methodological Rigor

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[Portfolio](#) • [GitHub](#)

IELTS: Overall Band 7.0 (no band less than 6.5) •  Bangladeshi

Research Objective: To develop statistically valid frameworks for computational social science, causal inference, and governance analytics with rigorous methodological foundations.

EDUCATION

Bachelor of Science in Computer Science & Engineering Rajshahi University of Engineering & Technology (RUET), Bangladesh CGPA: 3.08/4.00 • Thesis Grade: A+ • Seminar Grade: A+	Feb 2025
Higher Secondary Certificate (HSC) Pabna Cadet College, Bangladesh • Perfect GPA: 5.00/5.00 <i>Form Leader (led 60+ cadets); represented college at Spelling Bee, advancing to Divisional Round on three occasions; recipient of Best Speller Award at Pabna Cadet College (twice)</i>	2018
Secondary School Certificate (SSC) Pabna Cadet College, Bangladesh • Perfect GPA: 5.00/5.00	2016

RESEARCH INTERESTS

Computational Science; Statistical Validity in Machine Learning; Causal Inference; Governance Analytics; Measure-Theoretic Probability; Assumption Validation in Predictive Modeling; Development Economics; Robust Statistical Inference; Mathematical Methods in Data Science.

PUBLICATIONS

Accepted and Presented Paper:

- **Ahmed, S.J.** (2025). "Multi-Dimensional Statistical Similarity for Governance Classification: Beyond Arbitrary Thresholds in Comparative Politics." *6th Annual Paper Meet Electrical Engineering Division (APMEE 2025)*. Oral Presentation. **Received Best Research Paper Award.**

Accepted Papers (Forthcoming Presentation):

- **Ahmed, S.J.**, Islam Nahian, M.T., & Kwoshik, M.H.R. (2026). "RMA-BO: Regret-Minimizing Adaptive Bayesian Optimization." *Symposium on Photonics, Emerging Computational Technologies, Research & AI-Data Science (SPECTRA 2026)*, IEEE Photonics Society HSTU SBC. Track: Artificial Intelligence & Data Science. **Accepted for Presentation.**
- **Ahmed, S.J.**, Islam Nahian, M.T., & Kwoshik, M.H.R. (2026). "CF-EGAT: A Causal Fairness-Aware Equity Graph Attention Network for Country-Level Environmental Livability Classification." *Symposium on Photonics, Emerging Computational Technologies, Research & AI-Data Science (SPECTRA 2026)*, IEEE Photonics Society HSTU SBC. Track: Artificial Intelligence & Data Science. **Accepted for Presentation.**

Manuscripts Under Review (First-Author):

- **Ahmed, S.J.**, Islam Nahian, M.T., & Kwoshik, M.H.R. (2025). *Machine Learning for Crime Classification: A Fairness-Aware Approach to Class Imbalance.*
- **Ahmed, S.J.**, Islam Nahian, M.T., Kwoshik, M.H.R., & Nakib, F.N. (2025). *Environmental Livability Assessment via Adaptive Bootstrap-Retrained SHAP and Statistically-Constrained Pareto Counterfactuals: A Cross-National Analysis.*

RESEARCH EXPERIENCE

Researcher Royal Scientific Publications, Bangladesh	2026 – Present
• Conduct rigorous mathematical research and develop novel solutions addressing open problems in applied mathematics and computational science	

- Collaborate on peer-reviewed publications synthesizing mathematical theory with practical applications

Senior Researcher & Executive Committee Member

May 2022 – May 2024

Young Learners' Research Lab, Rajshahi University of Engineering & Technology (RUET)

- Conducted ML analysis, statistical inference, and numerical analysis for lab-wide initiatives; led hypothesis testing (Mann-Whitney U, Kolmogorov-Smirnov, Shapiro-Wilk) with Benjamini-Hochberg FDR correction across cross-disciplinary datasets
- Mentored 5+ junior researchers on Bayesian and causal inference; one mentee published their first conference paper as a direct outcome
- Established reproducible research standards and open science protocols adopted lab-wide
- *Awards:* Esteemed Alumni Award (2024); Certificate of Appreciation for Research Mentorship Excellence (2024)

Independent Researcher

2021 – Present

Department of Computer Science & Engineering, RUET

- *Annual Enterprise Survey EDA (2026):* One-Way ANOVA ($p < 0.001$) across all categorical variables on 55,620 enterprise records spanning 117 industries (New Zealand); hypothesis-driven statistical framework with professional visualizations; [GitHub](#)
- *Political Regime Classification:* Classified governance across 127 nations with 94% accuracy using hypothesis testing, FDR correction, and bootstrap validation; 95.4% consistency across 100 randomized configurations
- *Economic Development Classification:* 98.3% accuracy classifying 150+ countries on World Bank indicators via principled feature selection; [GitHub](#)
- *Crime Classification & Public Safety:* 93% accuracy on 6-class crime prediction (XGBoost) with spatial-temporal features and distributional assumption validation; [GitHub](#)

PROJECTS

StatsPro — AI-Powered Statistical Analysis Platform

2025 – 2026

[Live App](#) • Deployed on Streamlit Cloud

- Designed and deployed a full-stack statistical analysis web application that automates a complete data science workflow from raw CSV upload to professional report generation
- Integrated 10 rigorous statistical tests — normality (Shapiro-Wilk), correlation, chi-square, One-Way ANOVA, Mann-Whitney U, Levene, VIF, and OLS regression — with effect sizes and Benjamini-Hochberg FDR correction throughout
- Implemented automated preprocessing (missing data imputation, duplicate removal, outlier capping), 15+ visualizations (KDE histograms, Q-Q plots, heatmaps, pairplots), and instant data quality scoring
- Built ML Studio training 6 models (XGBoost, Random Forest, CatBoost, and others) with 5-fold cross-validation and live inference; integrated LLM-generated executive insights narrating findings like a senior data scientist

IMDb Transformer Sentiment Analysis — DistilBERT Fine-Tuning

2025

[GitHub](#)

- Fine-tuned pretrained **DistilBERT** on 10,000 IMDb reviews for binary sentiment classification; achieved **87.05% test accuracy** with 99.5% inference confidence on strong positive inputs
- Applied AdamW optimizer with linear warmup scheduler; training loss reduced from 0.4449 $\rightarrow$ 0.1615 over 3 epochs on T4 GPU (Google Colab)
- **Technical:** HuggingFace Transformers, PyTorch; benchmarked against classical ML baseline to quantify Transformer trade-offs on limited data

IMDb Sentiment Analysis — NLP & Classical ML Comparison

2025

[GitHub](#)

- Classified 50,000 IMDb movie reviews using TF-IDF vectorization; Logistic Regression achieved **89.76% accuracy** and 90.9% recall across stratified 80/20 split
- Compared three models (Logistic Regression, LinearSVC, Naive Bayes) under controlled evaluation; pipeline serialized as production-ready **.pkl** artifacts
- **Technical:** Unigram + bigram TF-IDF (10k features), scikit-learn; rigorous model selection via cross-validated performance metrics

TEACHING EXPERIENCE

Independent Tutor & Peer Mentor 2019 – Present
Self-employed, Dhaka, Bangladesh

- Provided free or low-cost tutoring in Physics, Chemistry, Higher Mathematics, ICT, and English to 50+ students from economically disadvantaged backgrounds; also mentored 10+ university peers in Vector Calculus, Statistics, Linear Algebra, and Machine Learning
- Mentored 5 students to achieve **perfect GPAs (5.00/5.00)** on national SSC and HSC board examinations
- Guided 4 students to gain admission to premier engineering universities (RUET, BUET, CUET)

HONORS AND AWARDS

2025	Best Research Paper Award, APMEE 2025 Conference
2024	Esteemed Alumni Award, Young Learners' Research Lab, RUET
2024	Certificate of Appreciation for Research Mentorship Excellence, RUET
2018	Perfect GPA (5.00/5.00), Higher Secondary Certificate (HSC)
2016	Perfect GPA (5.00/5.00), Secondary School Certificate (SSC)
2013, 09	Talentpool Scholarship (National Merit-based)
2016	General Scholarship (National Merit-based)
2013	Principal's Prize for Outstanding Academic Performance
2016–2018	Best Speller Award (×2), Pabna Cadet College

TECHNICAL SKILLS

Programming:	Python, C/C++, SQL, Git, Docker, Linux, LaTeX
Statistical Methods:	Probability Theory, Bayesian Inference, Causal Inference, Hypothesis Testing (Mann-Whitney U, Kolmogorov-Smirnov, Shapiro-Wilk), FDR Correction, Bootstrap
Machine Learning:	TensorFlow, Scikit-learn, XGBoost, Random Forest, Ensemble Methods, Feature Engineering, Cross-validation
Deep Learning:	TensorFlow, PyTorch, HuggingFace Transformers, DistilBERT Fine-Tuning, Neural Network Optimization
NLP:	TF-IDF Vectorization, Sentiment Analysis, Transformer Fine-Tuning
Mathematical Methods:	Numerical Analysis, Differential Equations, Linear Algebra, Optimization Theory
Data Visualization:	Matplotlib, Seaborn, Plotly
Research Tools:	Jupyter, Git/GitHub, Linux, LaTeX

COMMUNICATION & CREATIVE PURSUITS

Radio Jockey & Show Host 2020–2021
Radio Kothabondhu, Dhaka, Bangladesh

Creative Works

- **Published Poet:** Author of poetry collection “Sob Odvutire” (Bengali)
- **Lyricist, Composer & Singer:** Write lyrics, compose, and perform original songs.
- **Poetry Recitation:** First place in multiple competitions (2010–2013).

REFERENCES

Md. Nahiduzzaman
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